

NovaCred

Credit Application

Governance Analysis

Data Quality

Fairness & Bias

Privacy & Compliance

Team

Group TXB · Team 14 · DEGO 2606, Nova SBE

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Data loading, JSON flattening,
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structure

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Bias metrics, statistical testing,
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PII inventory, GDPR mapping, EU
AI Act classification,
pseudonymisation demo

Context & Objective

NovaCred is under regulatory inquiry for potential discrimination in lending — we act as the Data Governance Task Force

01

Audit Data Quality

Assessed 502 raw records across four quality dimensions. Resolved duplicate IDs, five-variant gender encoding, four date formats, and an undocumented income field merge. Output: 500 records, 34 columns.

02

Bias & Fairness Analysis

Assessed lending decisions for discriminatory patterns. Gender DI = 0.77 breaches the 0.80 four-fifths threshold. ZIP code is a strong gender proxy (AUC = 0.864). Age and intersectional disparities confirmed.

03

Privacy & Compliance

Catalogued PII across four risk categories. Identified 8 GDPR and EU AI Act gaps. Confirmed Annex III High-Risk classification. Demonstrated SHA-256 pseudonymisation and right-to-erasure simulation.

Data Quality — Findings & Remediation

Notebook 01 | João Marques Roque — Data Engineer

1.1 COMPLETENESS

Missing Values

processing_timestamp 87.6% · loan_purpose 90.0% · rejection_reason 58.2%
— structural gaps with GDPR audit trail implications

annual_salary → annual_income merge

Duplicate income field — annual_salary merged into annual_income, original column dropped, dtype corrected.

1.2 CONSISTENCY

Duplicate_id rows — 2 records (0.4%)

app_001 and app_042 each appeared twice with conflicting field values. Most complete record retained per ID.

Gender — 5 distinct values

Five variants across 113 records (M, F, Male, Female, blank). Standardised to Male/Female/Unknown — not imputed.

Date of Birth — 4 formats

Four formats coexisted across all 500 records. Each parsed individually via regex and standardised to ISO 8601.

1.3 VALIDITY

Invalid numeric values → NaN

Negative credit_history_months (2) and negative savings_balance (1) — domain impossibilities. Flagged and set to NaN; non-sensitive fields median-imputed.

Invalid email addresses → NaN (11, 2.2%)

Regex pattern applied (11 records flagged: missing @, invalid domain, blank). Coerced to NaN — PII not imputed.

ZIP code format violation (1 record, 0.2%)

One record with non-standard ZIP format — flagged for review, not silently overwritten.

1.4 ACCURACY & PLAUSIBILITY

Cross-field consistency — 0 violations

All approvals carry a positive approved_amount; all rejections carry a rejection_reason. No cross-field violations found.

Spending behaviour — strong integrity

No empty lists, missing categories, or negative amounts. All 15 spending categories present across all 500 records.

Governance-sensitive fields — NOT imputed

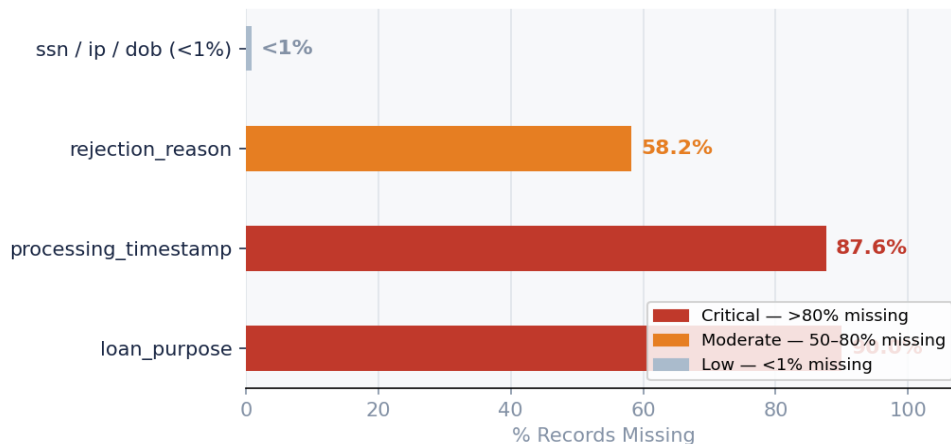
PII, decision outcomes, and rejection reasons were excluded from imputation

✓ Clean dataset: 500 rows × 34 columns | 502 → 500 after de-duplication | Primary key uniqueness restored | Governance-sensitive fields protected from imputation

Data Quality — Missingness & Outlier Analysis

Notebook 01 | João Marques Roque — Data Engineer

Structural Missingness by Field



loan_purpose — 90% missing

No record of why applicants applied. Cannot assess purpose-based risk. Governance gap: no collection policy documented.

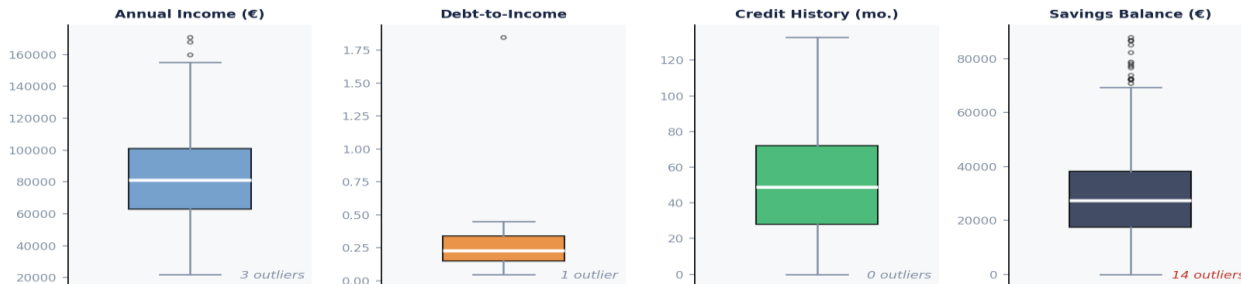
processing_timestamp — 87.6% missing

Cannot reconstruct the decision audit trail. GDPR Art. 5(1)(e) retention obligations cannot be verified.

rejection_reason — 58.2% missing

Over half of all rejected applicants received no documented reason. Of those with a reason, 170 cite algorithm_risk_score only.

IQR Outlier Detection — Key Financial Variables



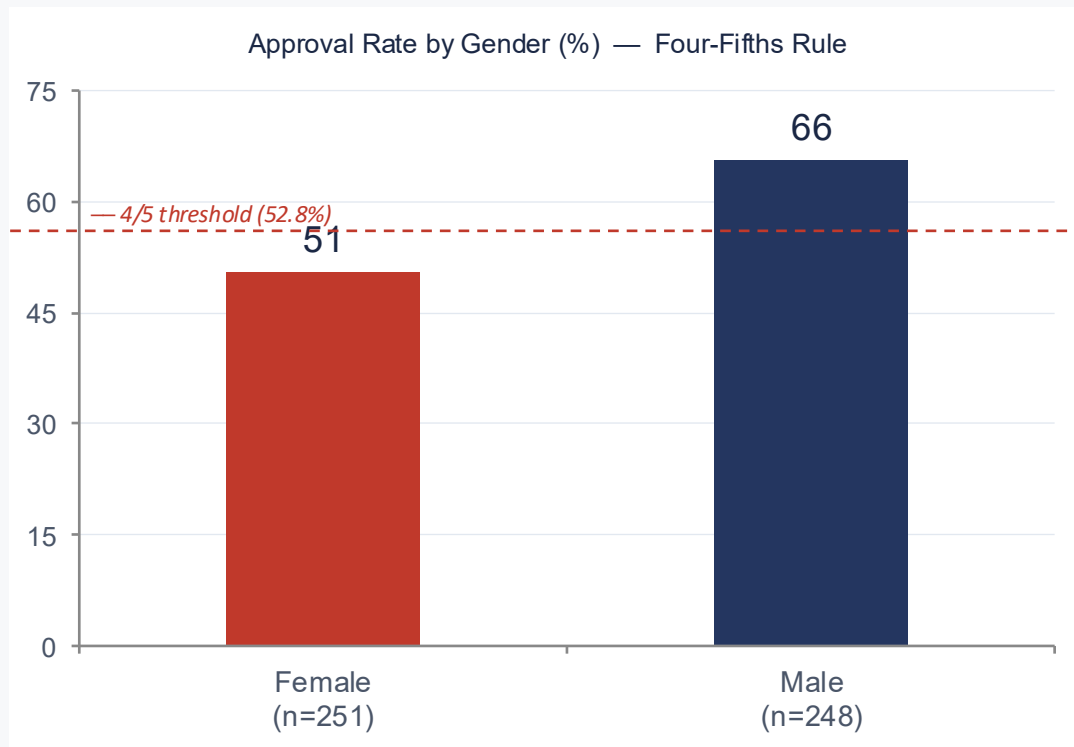
Data Quality — Issues Summary & Remediation

Notebook 01 | João Marques Roque — Data Engineer

#	Issue	Dimension	Field(s)	Count	%	Remediation
01	High missingness — notes	Completeness	notes	500	99.6 %	Dropped (≥90%); documented as optional context field
02	High missingness — annual_salary	Completeness	financials.annual_salary	497	99.0 %	Dropped (≥90%); merged into annual_income before exclusion
03	High missingness — loan_purpose	Completeness	loan_purpose	452	90.0 %	Dropped (≥90%); document collection policy for future audits
04	High missingness — processing_timestamp	Completeness	processing_timestamp	440	87.6 %	Retained; restore field — required for GDPR Art. 5(1)(e) audit trail
05	Duplicate application IDs (conflicting values)	Uniqueness	_id	2	0.4%	Retained most-complete record per _id
06	Gender encoded 5 inconsistent ways	Consistency	gender	113	22.5 %	Normalised to Male/Female/Unknown — not imputed
07	Date of birth in 4 different formats	Consistency	date_of_birth	500	100%	Parsed by regex; standardised to ISO 8601
08	Invalid email addresses	Validity	email	11	2.2%	Coerced to NaN — PII not imputed
09	Negative credit history / savings balance	Validity	credit_history_months, savings_balance	3	0.6%	Domain impossibilities → flagged and set to NaN
7 issues catalogued across 4 dimensions		2 critical gaps completeness — GDPR risk	Governance-sensitive fields never imputed		Clean output 500 rows × 34 columns	

Gender Disparate Impact

Notebook 02 | Maddalena Manfredi — Data Scientist



0.77

DI Ratio

Below 0.80 — legally significant

15.1 pp

Approval Gap

Absolute gap (male vs female)

V = 0.149

Effect Size (Cramér's V)

Small but statistically significant

p < 0.001

Chi-square test

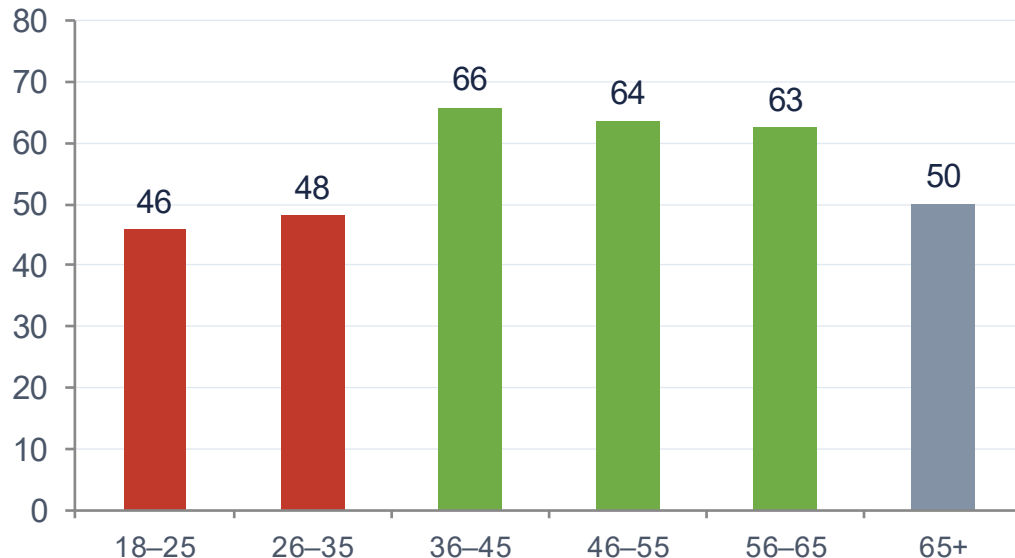
Pattern is real, not random

A DI ratio of 0.77 constitutes legally significant adverse impact under the four-fifths rule. Chi-square ($p = 0.0009$) and Cramér's $V = 0.149$ confirm the result is statistically robust.

Age-Based Discrimination Patterns

Notebook 02 | Maddalena Manfredi — Data Scientist

Approval Rate by Age Group (%)



The 18-25 and 26-35 cohorts record the lowest approval rates — both below 50% and significantly behind peak (36-45)

⚠ Intersectional finding (Young 18-35):

Female 38.1% vs Male 58.4% in the Young (18-35) cohort — a 20.3 pp gap, statistically significant ($p = 0.0088$). Aggregate metrics mask where disparity concentrates.

Statistical Evidence

Chi-square (18-35 vs 36-65)

$p = 0.0004$

Age-approval disparity is highly significant

Overall rate: Young vs Middle

47.8% vs 64.6%

Young (18-35) overall vs Middle (36-65) overall

Gender gap — Young cohort

20.3 pp

Female 38.1% vs Male 58.4% ($p = 0.0088$)

Gender gap — Middle cohort

10.6 pp

Female 59.1% vs Male 69.7% ($p = 0.0697$)

Intersectional Bias: Gender × Age

Notebook 02 | Maddalena Manfredi — Data Scientist

Age Group	Female Rate	n	Male Rate	n	Gender Gap (pp)	χ^2 test
Young (18–35)	38.1%	97	58.4%	89	20.3 X	p = 0.0088 *
Middle (36–65)	59.1%	149	69.7%	155	10.6	p = 0.0697
Senior (65+)	33.3%	3	66.7%	3	33.3	n < 6 — inconclusive

Detailed age breakdown (6 cohorts) — approval rate by group

45.8%

18–25 (n=24)

48.1%

26–35 (n=162)

65.8%

36–45 (n=164)

63.5%

46–55 (n=85)

62.5%

56–65 (n=56)

50.0%

65+ (n=6)

Key Finding:

The Young (18–35) cohort shows the only statistically significant gender gap (20.3 pp, p = 0.0088). The Middle (36–65) gap of 10.6 pp does not reach significance (p = 0.0697). Aggregate DI metrics mask where disparity is concentrated — subgroup monitoring is required for High-Risk AI under EU AI Act Annex III.

Proxy Variable Analysis

Notebook 02 | Maddalena Manfredi — Data Scientist

What is a Proxy Variable?

A variable that is not itself a protected attribute but correlates strongly with one (gender, age) and with the outcome. Including it in a model can reproduce discriminatory patterns indirectly.

Two combinations were tested as potential gender proxies:
zip_code × loan_approval and spending_behavior × income.

Legend:

	Gender proxy test		Statistical test
	Combined result		Not significant (—)

zip_code x loan_approval

 PROXY DETECTED

AUC (zip only): 0.864

AUC (approval only): 0.576

Mean CV AUC (combined):
0.849

Main findings:

zip_code alone has strong gender proxy — it predicts gender with AUC = 0.864, well above the 0.7 risk threshold. This reflects geographic segregation patterns encoded in residential areas.
loan_approval alone has weak predictive power — (AUC = 0.576) proxy effect is driven by geographic information. Must be excluded or transformed before model use.

spending_behavior x income

✓ NOT A PROXY

Not significant

Binomial test: $p > 0.05$

All pairs: n.s.

Spending habits × income tested as gender proxy. Gender distribution per spending category and income group examined via binomial tests. No statistically significant associations found (all $p > 0.05$) — spending behavior does not reliably predict gender in this dataset.

PII Identification & Classification

Notebook 03 | Lucas Galilei — Governance Officer

Direct Identifier

Uniquely identifies a person alone

e.g. full_name · ssn · email

Quasi-Identifier

Re-identification risk when combined

e.g. date_of_birth · zip_code

Technical ID

Identifies a device or session

e.g. ip_address

Non-Personal

System outputs — no direct link

e.g. loan_approved

5 High Risk

3 Medium

2 Low

Field	Category	Risk	Key Finding & Action
full_name, ssn, email	Direct Identifier	● High	Stored in plaintext — pseudonymise SSN; remove name and email from the analytical layer
ip_address	Technical Identifier	● High	No analytical value — delete from dataset (GDPR Art. 4(1))
gender	Protected Attribute	● High	DI = 0.77 confirmed — exclude from all model inputs
zip_code	Quasi-Identifier	● High	Gender proxy AUC = 0.864 — transform to NUTS-2 region or exclude
date_of_birth	Quasi-Identifier	● High	Exceeds data minimisation — generalise to age_group (GDPR Art. 5(1)(c))
annual_income	Quasi-Identifier	🕒 Med	Merged from annual_salary; reconciliation not documented
healthcare / gambling spend	Sensitive Behavioural	🕒 Med	Collection purpose undocumented — Art. 9 special-category risk applies
loan_approved	Decision Output	🕒 Med	Automated decision output — Art. 22 and Annex III obligations apply

Governance Gaps & Recommendations

Notebook 03 | Lucas Galilei — Governance Officer

01

No lawful basis or consent tracking

GDPR Art. 6 · Art. 7

Add `consent_timestamp`. Document lawful basis per processing activity prior to any model reuse.

02

PII stored in plaintext — no encryption

GDPR Art. 5(1)(f) · Art. 32

Apply SHA-256 pseudonymisation (demonstrated in Notebook 03). Encrypt data at rest; enforce role-based access.

03

No data retention policy or timestamps

GDPR Art. 5(1)(e) · Art. 17

Restore `processing_timestamp`. Define and automate a data retention and deletion schedule.

04

Behavioural data — no documented purpose

GDPR Art. 5(1)(b)(c) · Art. 22

Document purpose per spending category. Confirm applicant notification. Remove fields with no modelling value.

05

No audit trail — EU AI Act Art. 12 breach

EU AI Act Art. 12 · Art. 13

Introduce `decision_id`, `model_version`, and `decision_timestamp`. Maintain an append-only audit log.

06

169 automated rejections — no human review

GDPR Art. 22 · AI Act Art. 14

Require human review for all automated rejections. Establish a formal contestability pathway.

07

No explainability or appeal mechanism

GDPR Art. 22(3) · AI Act Art. 13

Deploy SHAP/LIME explanations for all decisions. Provide plain-language rationale and a documented appeal route.

08

High-Risk AI system — conformity to improve

AI Act Annex III · Art. 6 · Art. 10

Complete DPIA and EU AI Act conformity assessment. Appoint a Data Protection Officer. Deploy a fairness monitoring dashboard.

Key Findings & Conclusion

NovaCred requires fundamental governance redesign before its credit scoring system can be considered fair, compliant, and safe to deploy

01 Data Quality

- 502 → 500 records: app_001 and app_042 removed as conflicting duplicates
- Gender field had 5 encoding variants across 113 records — normalised
- Date of birth stored in 4 formats — standardised to ISO 8601
- annual_salary and annual_income were duplicate fields — silently merged
- 11 invalid emails, 2 negative credit months, 1 negative savings balance — all flagged

02 Algorithmic Bias

- Gender DI = 0.77 — breaches 0.80 four-fifths threshold ($p = 0.0009$)
- ZIP code: strong gender proxy (AUC = 0.864, mean CV AUC = 0.849)
- Young (18–35) vs Middle (36–65): $p = 0.0004$ — statistically significant
- Female 38.1% vs Male 58.4% in Young cohort — 20.3 pp gap ($p = 0.0088$)
- Only Young (18–35) gender gap statistically significant — Middle: $p = 0.0697$

03 Privacy & Compliance

- SSN, full name, email, IP all in plaintext — critical re-identification risk
- No lawful basis, consent mechanism, or retention policy in place
- 169 algorithmic rejections: no explanation, no review, no appeal pathway
- EU AI Act Annex III: High-Risk classification with zero conformity controls
- 8 GDPR / AI Act compliance gaps — each requiring concrete action

04 Recommended Actions

- Pseudonymise SSN; remove name, email, IP from analytical layer
- Exclude gender and ZIP from model inputs — both confirmed discriminatory
- Generalise date_of_birth to age_group prior to any further use
- Require human review for all rejections; add contestability mechanism
- Complete DPIA and AI Act conformity assessment; appoint DPO

NovaCred's credit scoring pipeline exhibits statistically confirmed gender and age disparities, stores PII in plaintext, and operates with zero compliance controls under the EU AI Act.

Recommendations Roadmap

15 remediation actions across three implementation horizons — sequenced by legal urgency

